

"Many Forms"

Polymorphism occurs when different classes are related via inheritance. It allows a single method to perform different actions based on the specific object that is calling it.

METHOD OVERRIDING

1. Defining the Logic

Subclasses provide their own implementation of a method defined in the parent class.

```
class Animal {  
    public void makeSound() { System.out.println("The animal makes a sound"); }  
}  
  
class Pig extends Animal {  
    public void makeSound() { System.out.println("The pig says: wee wee"); }  
}
```

2. Unified Reference

You can use a parent class variable to store child class objects. Java decides which method to run at **runtime**.

```
Animal myAnimal = new Animal(); // Parent reference, Parent object
Animal myPig = new Pig();        // Parent reference, Child object

myAnimal.makeSound(); // Output: Parent sound
myPig.makeSound();    // Output: Pig sound!
```

Core Advantages

- **Flexibility:** Write code that works with a Parent type, and it will automatically support all Child types.
- **Scalability:** Add new subclasses without modifying the main logic of your application.
- **Reusability:** Rely on existing parent attributes while customizing specific child behaviors.